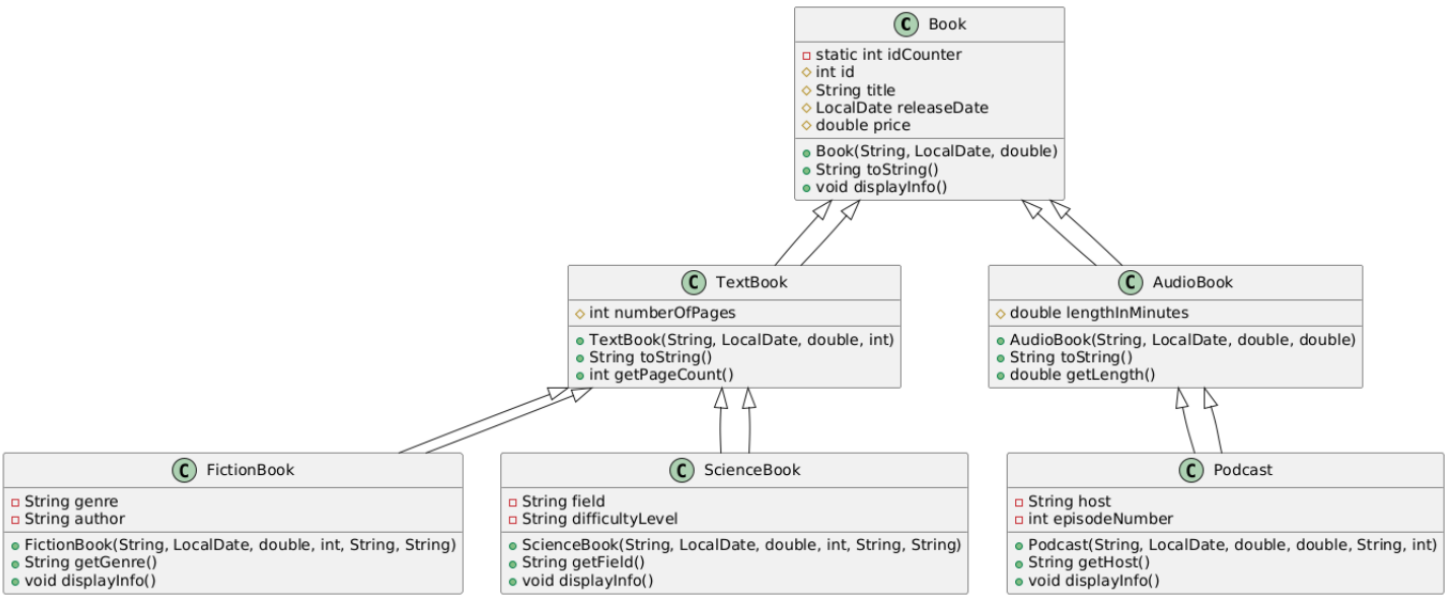
Lab 05

Assume an online Bookstore that sells two types of Books: Textbooks and audio Books. For a Text book, we need to keep track of the ID (int), title (String), release Date (java.time.LocalDate),price(double), and number of pages(int). For an Audio Book, we need to keep track of the ID (int), title (String), release Date (java.time.LocalDate), price (double), and length in minutes (double). The ID is just a serial number initialized at construction time. This means that the first Book object constructed should have id= 1 and the second one id= 2, and so on. Textbook has two types of books, FictionBook and ScienceBook. Audiobooks can be of a specific type of podcast. Provided is a UML



The following code provides the implementation of classes in the UML

```
import java.time.LocalDate;


class Book {

   private static int idCounter = 1;

   protected int id;
```

```
    protected String title;
    protected LocalDate releaseDate;
    protected double price;

    public Book(String title, LocalDate releaseDate, double price) {
        this.id = idCounter++;
        this.title = title;
        this.releaseDate = releaseDate;
        this.price = price;
    }

    @Override
    public String toString() {
        return "Book ID: " + id + ", Title: " + title + ", Release Date: " + releaseDate + ", Price: $" +
price;
    }

    public void displayInfo() {
        System.out.println("Book Information:");
        System.out.println("ID: " + id);
        System.out.println("Title: " + title);
        System.out.println("Release Date: " + releaseDate);
        System.out.println("Price: $" + price);
        System.out.println("------------------------");
    }
}
```

```
class TextBook extends Book {
   protected int numberOfPages;

   public TextBook(String title, LocalDate releaseDate, double price, int numberOfPages) {
      super(title, releaseDate, price);
      this.numberOfPages = numberOfPages;
   }

   @Override
   public String toString() {
            System.out.println("MethodID: TS11");
      return super.toString() + ", Pages: " + numberOfPages;
   }


   public int getPageCount() {
      return numberOfPages;
   }
}

class FictionBook extends TextBook {
   private String genre;
   private String author;
```

```
  public FictionBook(String title, LocalDate releaseDate, double price, int numberOfPages,
String genre, String author) {
    super(title, releaseDate, price, numberOfPages);
    this.genre = genre;
    this.author = author;
  }
        @Override
  public void displayInfo() {
        System.out.println("MethodID: DI21");
    super.displayInfo();
    System.out.println("Subtype: FictionBook");
    System.out.println("Genre: " + genre);
    System.out.println("Author: " + author);
    System.out.println("========================");
  }

  public String getGenre() {
    return genre;
  }
}

class ScienceBook extends TextBook {
  private String field;
  private String difficultyLevel;

  public ScienceBook(String title, LocalDate releaseDate, double price, int
numberOfPages, String field, String difficultyLevel) {
```

```
    super(title, releaseDate, price, numberOfPages);
    this.field = field;
    this.difficultyLevel = difficultyLevel;
  }
        @Override
  public void displayInfo() {
        System.out.println("MethodID: DI22");
    super.displayInfo();
    System.out.println("Subtype: ScienceBook");
    System.out.println("Field: " + field);
    System.out.println("Difficulty Level: " + difficultyLevel);
    System.out.println("========================");
  }

  public String getField() {
    return field;
  }
}

class AudioBook extends Book {
  protected double lengthInMinutes;

  public AudioBook(String title, LocalDate releaseDate, double price, double
lengthInMinutes) {
    super(title, releaseDate, price);
    this.lengthInMinutes = lengthInMinutes;
```

```
    }

    @Override
    public String toString() {
            System.out.println("MethodID: TS12");
        return super.toString() + ", Length: " + lengthInMinutes + " minutes";
    }

    public double getLength() {
        return lengthInMinutes;
    }
}

class Podcast extends AudioBook {
    private String host;
    private int episodeNumber;

    public Podcast(String title, LocalDate releaseDate, double price, double lengthInMinutes,
String host, int episodeNumber) {
        super(title, releaseDate, price, lengthInMinutes);
        this.host = host;
        this.episodeNumber = episodeNumber;
    }
            @Override
    public void displayInfo() {
            System.out.println("MethodID: DI23");
```

```
        super.displayInfo();
    System.out.println("Subtype: Podcast");
    System.out.println("Host: " + host);
    System.out.println("Episode Number: " + episodeNumber);
    System.out.println("========================");
  }

  public String getHost() {
    return host;
  }

        public int getEpisodeNumber() {
                return episodeNumber;
        }
}
```

Write a program that

1. **Creates an array** of different book types (Book, TextBook, AudioBook, FictionBook, ScienceBook, Podcast).
    - TextBook: "Java Programming", LocalDate.of(2021, 5, 10), 45.99, 500
    - AudioBook: "Learn Python Fast", LocalDate.of(2022, 7, 20), 15.99, 120
    - FictionBook: "Harry Potter", LocalDate.of(1997, 6, 26), 39.99, 400, "Fantasy", "J.K. Rowling"
    - ScienceBook("Physics for Beginners", LocalDate.of(2020, 9, 15), 29.99, 350, "Physics", "Advanced"
    - Podcast :"Tech Talk", LocalDate.of(2023, 3, 25), 5.99, 45, "John Doe", 10

2. **Displays info** for all books. Which method is executed every time displayinfo() is called?

3. **Prints the following specific details**:
    a. host and episode number for **Podcasts**.
    b. number of pages for **TextBooks**.
4. **Print. toString()** output for all books. Which method is executed every time toString() is called?
5. **Delete method DisplayInfo()** from class **book.** Do you expect your code to compile? Does your code compile? Discuss!

6. Delete method **toString()** from class **book**. Do you expect your code to compile? Does your code compile? Discuss!